

In order to use the third expression in Equation (1.71) we first need to obtain the eigendata (i.e., the eigenvalues and eigenvectors) of the A matrix. These are found as

$$\begin{aligned}\lambda(A) &= \{0, 0\} \\ v(A) &= \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}\end{aligned}\quad (1.80)$$

This shows that

$$\begin{aligned}\hat{A} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ Q &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\end{aligned}\quad (1.81)$$

Note that in this simple example A is already in Jordan form, so $\hat{A} = A$ and $Q = I$. The third expression in Equation (1.71) therefore gives

$$\begin{aligned}e^{At} &= Qe^{\hat{A}t}Q^{-1} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}\end{aligned}\quad (1.82)$$

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1.3 NONLINEAR SYSTEMS

The discussion of linear systems in the preceding section is a bit optimistic, because in reality linear systems do not exist. Real systems always have some nonlinearities. Even a simple resistor is ultimately nonlinear if we apply a large enough voltage across it. However, we often model a resistor with the simple linear equation $V = IR$ because this equation accurately describes the operation of the resistor over a wide operating range. So even though linear systems do not exist in the real world, linear systems theory is still a valuable tool for dealing with nonlinear systems.

The general form of a continuous-time nonlinear system can be written as

$$\begin{aligned}\dot{x} &= f(x, u, w) \\ y &= h(x, v)\end{aligned}\quad (1.83)$$

where $f(\cdot)$ and $h(\cdot)$ are arbitrary vector-valued functions. We use w to indicate process noise, and v to indicate measurement noise. If $f(\cdot)$ and $h(\cdot)$ are explicit functions of t then the system is time-varying. Otherwise, the system is time-invariant. If $f(x, u, w) = Ax + Bu + w$, and $h(x, v) = Hx + v$, then the system is linear [compare with Equation (1.67)]. Otherwise, the system is nonlinear.

In order to apply tools from linear systems theory to nonlinear systems, we need to linearize the nonlinear system. In other words, we need to find a linear system

that is approximately equal to the nonlinear system. To see how this is done, let us start with a nonlinear vector function $f(\cdot)$ of a scalar x . We expand $f(x)$ in a Taylor series around some nominal operating point (also called a linearization point) $x = \bar{x}$, defining $\tilde{x} = x - \bar{x}$:

$$f(x) = f(\bar{x}) + \left. \frac{\partial f}{\partial x} \right|_{\bar{x}} \tilde{x} + \frac{1}{2!} \left. \frac{\partial^2 f}{\partial x^2} \right|_{\bar{x}} \tilde{x}^2 + \frac{1}{3!} \left. \frac{\partial^3 f}{\partial x^3} \right|_{\bar{x}} \tilde{x}^3 + \dots \quad (1.84)$$

Now suppose that x is a 2×1 vector. This implies that $f(x)$ is a nonlinear function of two independent variables x_1 and x_2 . The Taylor series expansion of $f(x)$ becomes

$$\begin{aligned} f(x) = & f(\bar{x}) + \left. \frac{\partial f}{\partial x_1} \right|_{\bar{x}} \tilde{x}_1 + \left. \frac{\partial f}{\partial x_2} \right|_{\bar{x}} \tilde{x}_2 + \\ & \frac{1}{2!} \left(\left. \frac{\partial^2 f}{\partial x_1^2} \right|_{\bar{x}} \tilde{x}_1^2 + \left. \frac{\partial^2 f}{\partial x_2^2} \right|_{\bar{x}} \tilde{x}_2^2 + 2 \left. \frac{\partial^2 f}{\partial x_1 x_2} \right|_{\bar{x}} \tilde{x}_1 \tilde{x}_2 \right) + \\ & \frac{1}{3!} \left(\left. \frac{\partial^3 f}{\partial x_1^3} \right|_{\bar{x}} \tilde{x}_1^3 + \left. \frac{\partial^3 f}{\partial x_2^3} \right|_{\bar{x}} \tilde{x}_2^3 + 3 \left. \frac{\partial^3 f}{\partial x_1^2 x_2} \right|_{\bar{x}} \tilde{x}_1^2 \tilde{x}_2 + 3 \left. \frac{\partial^3 f}{\partial x_1 x_2^2} \right|_{\bar{x}} \tilde{x}_1 \tilde{x}_2^2 \right) + \dots \end{aligned} \quad (1.85)$$

This can be written more compactly as

$$\begin{aligned} f(x) = & f(\bar{x}) + \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \tilde{x}_2 \frac{\partial}{\partial x_2} \right) f \Big|_{\bar{x}} + \frac{1}{2!} \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \tilde{x}_2 \frac{\partial}{\partial x_2} \right)^2 f \Big|_{\bar{x}} + \\ & \frac{1}{3!} \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \tilde{x}_2 \frac{\partial}{\partial x_2} \right)^3 f \Big|_{\bar{x}} + \dots \end{aligned} \quad (1.86)$$

Extending this to the general case in which x is an $n \times 1$ vector, we see that any continuous vector-valued function $f(x)$ can be expanded in a Taylor series as

$$\begin{aligned} f(x) = & f(\bar{x}) + \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \dots + \tilde{x}_n \frac{\partial}{\partial x_n} \right) f \Big|_{\bar{x}} + \\ & \frac{1}{2!} \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \dots + \tilde{x}_n \frac{\partial}{\partial x_n} \right)^2 f \Big|_{\bar{x}} + \\ & \frac{1}{3!} \left(\tilde{x}_1 \frac{\partial}{\partial x_1} + \dots + \tilde{x}_n \frac{\partial}{\partial x_n} \right)^3 f \Big|_{\bar{x}} + \dots \end{aligned} \quad (1.87)$$

Now we define the operation $D_{\tilde{x}}^k f$ as

$$D_{\tilde{x}}^k f = \left(\sum_{i=1}^n \tilde{x}_i \frac{\partial}{\partial x_i} \right)^k f(x) \Big|_{\bar{x}} \quad (1.88)$$

Using this definition we write the Taylor series expansion of $f(x)$ as

$$f(x) = f(\bar{x}) + D_{\tilde{x}} f + \frac{1}{2!} D_{\tilde{x}}^2 f + \frac{1}{3!} D_{\tilde{x}}^3 f + \dots \quad (1.89)$$

If the nonlinear function $f(x)$ is "sufficiently smooth," then high-order derivatives of $f(x)$ should be "somewhat small." Also, if $f(x)$ is expanded around a point such

that x is “close” to $\bar{x}$, then $\tilde{x}$ will be “small” and the higher powers of $\tilde{x}$ in Equation (1.89) will be “small.” Finally, the higher-order derivatives in the Taylor series expansion of Equation (1.89) are divided by increasingly large factorials, which further diminishes the magnitude of the higher-order terms in Equation (1.89). This justifies the approximation

$$\begin{aligned} f(x) &\approx f(\bar{x}) + D_{\bar{x}}f \\ &\approx f(\bar{x}) + \left. \frac{\partial f}{\partial x} \right|_{\bar{x}} \tilde{x} \\ &\approx f(\bar{x}) + A\tilde{x} \end{aligned} \quad (1.90)$$

where A is the matrix defined by the above equation.

Returning to our nonlinear system equations in Equation (1.83), we can expand the nonlinear system equation $f(x, u, w)$ around the nominal operating point $(\bar{x}, \bar{u}, \bar{w})$. We then obtain a linear system approximation as follows.

$$\begin{aligned} \dot{x} &= f(x, u, w) \\ &\approx f(\bar{x}, \bar{u}, \bar{w}) + \left. \frac{\partial f}{\partial x} \right|_0 (x - \bar{x}) + \left. \frac{\partial f}{\partial u} \right|_0 (u - \bar{u}) + \left. \frac{\partial f}{\partial w} \right|_0 (w - \bar{w}) \\ &= \dot{\bar{x}} + A\tilde{x} + B\tilde{u} + L\tilde{w} \end{aligned} \quad (1.91)$$

where the 0 subscript means that the function is evaluated at the nominal point $(\bar{x}, \bar{u}, \bar{w})$, and A , B , and L are defined by the above equations. Subtracting $\dot{\bar{x}}$ from both sides of Equation (1.91) gives

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u} + L\tilde{w} \quad (1.92)$$

Since w is noise, we will set $\bar{w} = 0$ so that $\tilde{w} = w$ and we obtain

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u} + Lw \quad (1.93)$$

We see that we have a linear equation for $\dot{\tilde{x}}$ in terms of $\tilde{x}$, $\tilde{u}$, and w . We have a linear equation for the deviations of the state and control from their nominal values. As long as the deviations remain small, the linearization will be accurate and the linear equation will accurately describe deviations of x from its nominal value $\bar{x}$.

In a similar manner we can expand the nonlinear measurement equation given by Equation (1.83) around a nominal operating point $x = \bar{x}$ and $v = \bar{v} = 0$. This results in the linearized measurement equation

$$\begin{aligned} \tilde{y} &= \left. \frac{\partial h}{\partial x} \right|_0 \tilde{x} + \left. \frac{\partial h}{\partial v} \right|_0 \tilde{v} \\ &= C\tilde{x} + Dv \end{aligned} \quad (1.94)$$

where C and D are defined by the above equation. Equations (1.93) and (1.94) comprise a linear system that describes the deviations of the state and output from their nominal values. Recall that the tilde quantities in Equations (1.93) and (1.94) are defined as

$$\begin{aligned} \tilde{x} &= x - \bar{x} \\ \tilde{u} &= u - \bar{u} \\ \tilde{y} &= y - \bar{y} \end{aligned} \quad (1.95)$$

■ EXAMPLE 1.4

Consider the following model for a two-phase permanent magnet synchronous motor:

$$\begin{aligned}
 \dot{i}_a &= -\frac{R}{L}i_a + \frac{\omega\lambda}{L}\sin\theta + \frac{u_a}{L} \\
 \dot{i}_b &= -\frac{R}{L}i_b - \frac{\omega\lambda}{L}\cos\theta + \frac{u_b}{L} \\
 \dot{\omega} &= \frac{-3\lambda}{2J}i_a\sin\theta + \frac{3\lambda}{2J}i_b\cos\theta - \frac{F\omega}{J} - \frac{T_l}{J} \\
 \dot{\theta} &= \omega
 \end{aligned} \tag{1.96}$$

where i_a and i_b are the currents through the two windings, R and L are the resistance and inductance of the windings, θ and ω are the angular position and velocity of the rotor, λ is the flux constant of the motor, u_a and u_b are the voltages applied across the two windings, J is the moment of inertia of the rotor and its load, F is the viscous friction of the rotor, and T_l is the load torque. The time variable does not explicitly appear on the right side of the above equation, so this is a time-invariant system. However, the system is highly nonlinear and we therefore cannot directly use any linear systems tools for control or estimation. However, if we linearize the system around a nominal (possibly time-varying) operating point then we can use linear system tools for control and estimation. We start by defining a state vector as $x = [i_a \ i_b \ \omega \ \theta]^T$. With this definition we write

$$\begin{aligned}
 \dot{x} &= [\dot{x}_1 \ \dot{x}_2 \ \dot{x}_3 \ \dot{x}_4]^T \\
 &= f(x, u) \\
 &= \begin{bmatrix} -\frac{R}{L}x_1 + \frac{x_3\lambda}{L}\sin x_4 + \frac{u_a}{L} \\ -\frac{R}{L}x_2 - \frac{x_3\lambda}{L}\cos x_4 + \frac{u_b}{L} \\ \frac{-3\lambda}{2J}x_1\sin x_4 + \frac{3\lambda}{2J}x_2\cos x_4 - \frac{F x_3}{J} - \frac{T_l}{J} \\ x_3 \end{bmatrix}
 \end{aligned} \tag{1.97}$$

We linearize the system equation by taking the partial derivative of $f(x, u)$ with respect to x and u to obtain

$$\begin{aligned}
 A &= \frac{\partial f}{\partial x} \\
 &= \begin{bmatrix} -R/L & 0 & \lambda s_4/L & x_3\lambda c_4/L \\ 0 & -R/L & -\lambda c_4/L & x_3\lambda s_4/L \\ -3\lambda s_4/2J & 3\lambda c_4/2J & -F/J & -3\lambda(x_1c_4 + x_2s_4)/2J \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
 B &= \frac{\partial f}{\partial u} \\
 &= \begin{bmatrix} 1/L & 0 \\ 0 & 1/L \\ 0 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned} \tag{1.98}$$